

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 2: Communication

Subtopic 2.1: Networks including the Internet

Concept 2.1.4: Network Topologies (Bus, Star, Mesh, Hybrid)

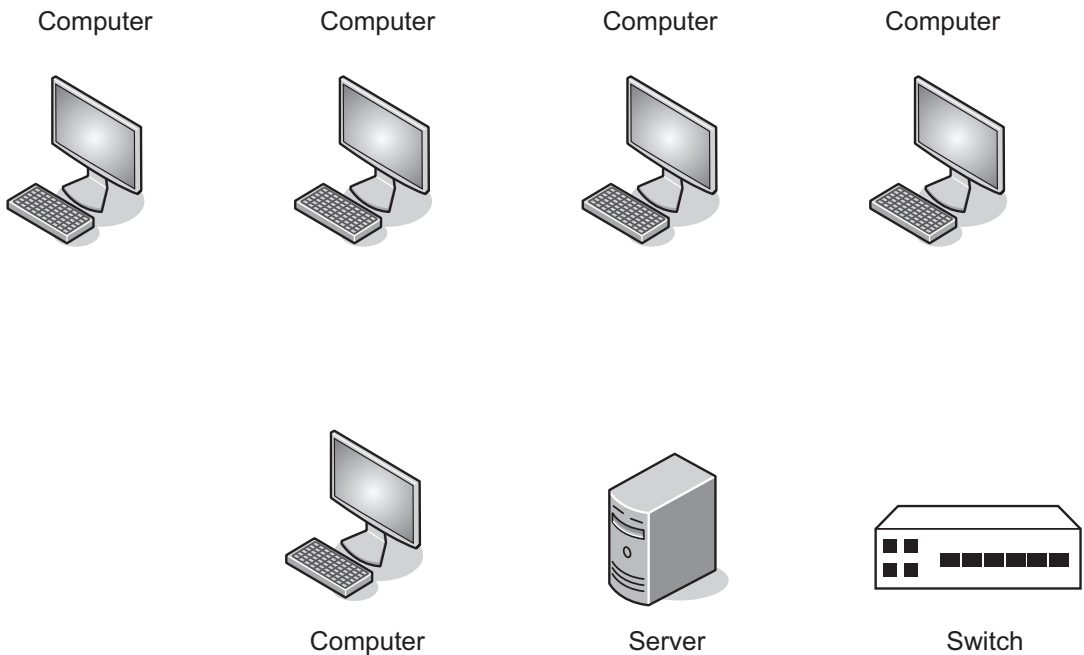
Total Questions: 12

Mark Schemes begin on Page 8

Question 1 | Source: 9618_w25_qp_12 (Q5.b)

5 A local area network (LAN) has five computers, one switch and one server.

(b) Complete the following diagram to show how these devices are connected in a star topology.



[2]

Question 2 | Source: 9618_w25_qp_12 (Q5.c)

5 A local area network (LAN) has five computers, one switch and one server.

(c) Describe how packets are transmitted between two hosts using a star topology.

.....

.....

.....

.....

.....

..... [3]

Question 3 | Source: 9618_s25_qp_12 (Q6.d)

6 A company has multiple sites in different cities.
The company has drivers who deliver products to customers. Each driver can connect to the company's WAN (Wide Area Network) whilst out of the office using a smartphone.

- (d) The main office for the company has four wired computers, one server, two printers, one central switch and a device providing a single point of access for the internet. The network is set up as a star topology.

Draw the topology for the main office network. Label all devices.

[4]

Question 4 | Source: 9618_s24_qp_11 (Q8.a)

- 8 A business is creating a local area network (LAN) in its office.

- (a) The business is deciding which topology to use.

Tick (✓) **one or more** boxes in each row to identify the topology, or topologies, each statement describes.

Statement	Bus	Star	Mesh
all devices connect to one central device			
all devices connect to a central cable			
multiple paths for the packets to travel along			
robust against damage because if any line fails, the rest of the network retains full functionality			
most likely to lose data through collisions			

[5]

Question 5 | Source: 9618_s24_qp_13 (Q5.b)

- 5 A multimedia design company has an office with a LAN (local area network). The LAN can have up to 20 devices connected with cables and other devices connected using wireless access.

- (b) Part of the internal structure of the wired LAN is a star topology.

Explain how packets are transmitted between two devices in a star topology.

.....

.....

.....

.....

[2]

Question 6 | Source: 9618_w23_qp_11 (Q2.d)

- 2 A school has a Local Area Network (LAN).

- (d) A new classroom is being set up with 20 computers and a switch.

Explain **one** advantage of implementing a star topology instead of a bus topology in the new classroom.

.....

.....

.....

.....

[2]

Question 7 | Source: 9618_w23_qp_12 (Q7.b.i)

7 A Local Area Network (LAN) contains four devices:

- a router
- two laptop computers
- a server.

(b) (i) The LAN is set up as a star topology.

Draw a diagram of the topology of the LAN.



[2]

Question 8 | Source: 9618_s23_qp_12 (Q1.c)

1 A company has a LAN (local area network).

(c) One of the company's offices has one server and four computers connected in a star topology.

Draw a diagram to show the layout of the office's star topology.

[2]

Question 9 | Source: 9618_s23_qp_13 (Q2.b.i)

- 2 A university has two sites. Each site has several computer rooms. The computers are all connected as a WAN (wide area network).
- (b) The network uses different topologies in different areas of the sites. In one building there are five computers connected in a mesh topology.

(i) Describe what is meant by a mesh topology.

.....

.....

.....

..... [2]

Question 10 | Source: 9618_s23_qp_13 (Q2.b.ii)

- 2 A university has two sites. Each site has several computer rooms. The computers are all connected as a WAN (wide area network).
- (b) The network uses different topologies in different areas of the sites. In one building there are five computers connected in a mesh topology.

(ii) Give **two** advantages of using a mesh topology instead of a bus topology.

1

.....

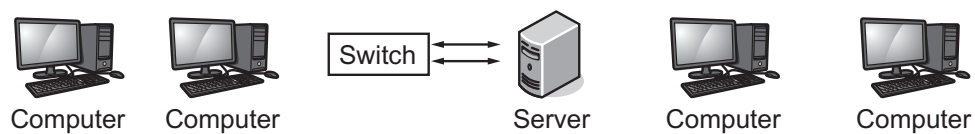
2

..... [2]

Question 11 | Source: 9618_w22_qp_12 (Q10.a)

- 10 A Local Area Network (LAN) consists of four computers, one server and a switch.
- The LAN uses a star topology.

- (a) Complete the following diagram to show how the hardware is connected.



[1]

Question 12 | Source: 9618_w21_qp_12 (Q3.b.i)

- 3 Andy likes to play computer games.

- (b) Andy plays some of the computer games over the internet. He has several devices that connect wirelessly to the router in his house.

- (i) Identify the topology of Andy's home network. Justify your choice.

Topology

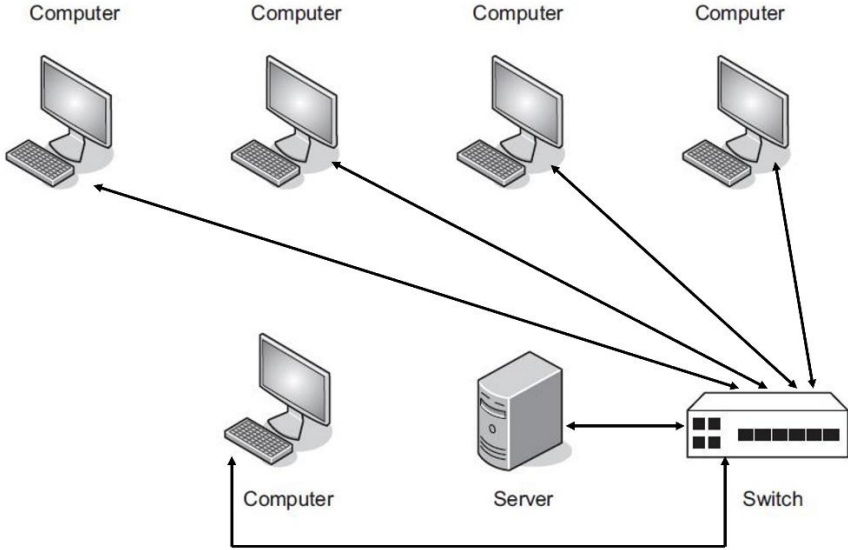
Justification

.....

[2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q5.b)

5(b)	1 mark per bullet point, max 2 marks - All computers directly connected to switch and no other connections - Switch connected directly to server  <pre> graph TD C1[Computer] --> S[Switch] C2[Computer] --> S C3[Computer] --> S C4[Computer] --> S C5[Computer] --> S S <--> SR[Server] </pre>	2
------	--	----------

Mark Scheme for Question 2 | Source: 9618_w25_ms_12 (Q5.c)

5(c)	1 mark per bullet point, max 3 marks - The sending device sends the packets to the switch - The switch checks the destination address on each packet ... and sends them only to the intended receiver	3
------	---	----------

Mark Scheme for Question 3 | Source: 9618_s25_ms_12 (Q6.d)

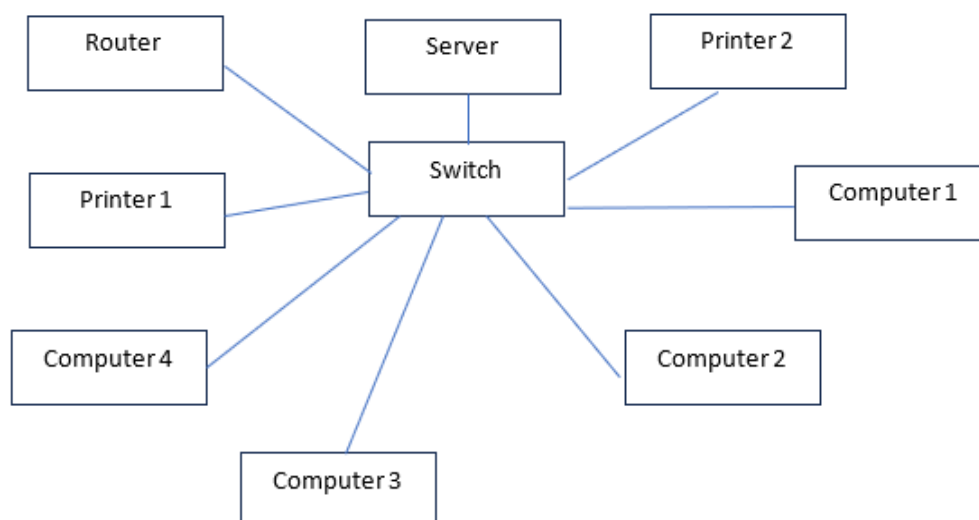
6(d)

1 mark each

4

- Server directly connected to the switch
- Four labelled computers all connected directly to the switch
- Two printers each directly connected to the switch / server / a computer
- Router / modem directly connected to the switch / server for internet access

e.g. Example only



Mark Scheme for Question 4 | Source: 9618_s24_ms_11 (Q8.a)

8(a)

1 mark for each correct row:

5

Statement	Bus	Star	Mesh
all devices connect to one central device		✓	
all devices connect to a central cable	✓		
multiple paths for the packets to travel			✓
robust against damage because if any line fails, the rest of the network retains full functionality		✓	✓
most likely to lose data through collisions	✓		

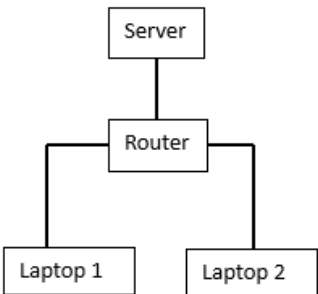
Mark Scheme for Question 5 | Source: 9618_s24_ms_13 (Q5.b)

5(b)	1 mark each: - Sending computer transmits packets directly to switch/router/central device - Switch/router/central device checks destination address of packet and forwards directly to that device	2
------	--	----------

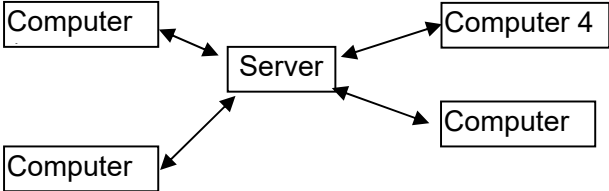
Mark Scheme for Question 6 | Source: 9618_w23_ms_11 (Q2.d)

2(d)	1 mark for each advantage and 1 mark for valid corresponding expansion - Star topology is more resilient to faults ...because there is no single cable and leads to less disruption to teaching - Higher performance as fewer collisions ...because each device in the classroom is only connected to the switch - Easier to add new nodes ...because each device in the classroom connects directly to the switch - Easier to fault find compared to bus topology	2
------	--	----------

Mark Scheme for Question 7 | Source: 9618_w23_ms_12 (Q7.b.i)

7(b)(i)	1 mark for server only connected to router 1 mark for two laptop computers connected only to router  <pre> graph TD Server[Server] --- Router[Router] Router --- Laptop1[Laptop 1] Router --- Laptop2[Laptop 2] </pre>	2
---------	---	----------

Mark Scheme for Question 8 | Source: 9618_s23_ms_12 (Q1.c)

1(c)	1 mark each to max 2 - Each computer directly connected only to the server <u>all</u> components correctly labelled 	2
------	---	----------

Mark Scheme for Question 9 | Source: 9618_s23_ms_13 (Q2.b.i)

2(b)(i)	1 mark each to max 2 - All computers are connected to at least one other device - There are multiple routes between devices - The computers can act as relays, passing packets on towards the final destination	2
---------	--	----------

Mark Scheme for Question 10 | Source: 9618_s23_ms_13 (Q2.b.ii)

2(b)(ii)	1 mark each to max 2 - If one line goes down there are more routes available - Improved security as not using one main line - No/fewer collisions - New nodes can be added without interruption or interfering with other nodes - More secure because data is sent over a dedicated connection	2
----------	---	----------

Mark Scheme for Question 11 | Source: 9618_w22_ms_12 (Q10.a)

10(a)	all four computers directly connected to the <u>switch</u> and no other connections.	1
-------	--	----------

Mark Scheme for Question 12 | Source: 9618_w21_ms_12 (Q3.b.i)

3(b)(i)	1 mark for identification of star topology 1 mark for justification Devices are connected directly to the <u>router</u> independently // all devices are only connected to the <u>router</u>	2
---------	---	----------